**NPTEL Online Certification Course**
**Fundamentals of Artificial Intelligence**
**MOOCs; July - Dec 2026**
**Indian Institute of Technology Guwahati**
**Assignment No. 6: Detailed Solution**

10 Marks

Each question carries 01 Mark each.

Q1. Which of the following statements are correct regarding entailment in First-Order Logic?

I. There exists an algorithm that halts and answers YES whenever $KB \models \alpha$.
II. There is no algorithm that always halts and answers NO whenever $KB \models/\alpha$.

Choose the correct option.
A. Both I and II
B. I only
C. II only
D. Neither I nor II
**Correct option: (A)**

Q2. In first order logic, if a set of sentences is _____ then resolution will always be able to derive a contradiction.
A. unsatisfiable
B. conjunction of literals
C. disjunction of literals
D. satisfiable

**Correct option: (A)**

Q3. A clause $C_1$ subsumes a clause $C_2$ if there exists a substitution $s$ such that
A. $C_1 s \subseteq C_2$
B. $C_2 s \subseteq C_1$
C. $C_1 \subseteq C_2 s$
D. $C_1 s = C_2$
**Correct option: (A)**

Q4. Which inference step is illustrated below?

$\forall x\ \text{Likes}(x, \text{IceCream})$ Therefore $\text{Likes}(\text{Rahul}, \text{IceCream})$

A. Existential Generalization
B. Universal Instantiation

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C. Existential Instantiation
D. Resolution

**Correct option: (B)**

Q5. The _____ strategy restricts each resolution step so that one parent clause must be either a member of the original clause set, or an ancestor of the other parent clause in the derivation.

A. Unit resolution
B. Input resolution
C. Linear resolution
D. Set-of-support

**Correct option:** (C)

**From Q6 to Q10**

Consider the following representation in First Order Logic:

1. A horse that is registered for today's race is not a thoroughbred.
2. Every horse registered for today's race has won a race this year.
3. A horse that has won a race this year is not a thoroughbred.

P(x): x is registered for today's race.
Q(x): x is a thoroughbred.
R(x): x has won a race this year.
U = all horses

Q6. In first-order logic, each variable refers to some object in a set called the _____.
A. domain of discourse
B. knowledge base
C. interpretation
D. semantics
**Correct option: (A)**

Q7. Translate the following English sentence to FOL.

Every horse that is registered for today's race is not a thoroughbred.

A. $\forall x\,[P(x) \vee Q(x)]$
B. $\forall x[P(x) \rightarrow \neg Q(x)]$
C. $\exists x[P(x) \wedge \neg Q(x)]$
D. $\forall x\,[P(x) \rightarrow Q(x)]$
**Correct option: (B)**

Q8. Translate the following English sentence to First Order Logic.

Every horse registered for today's race has won a race this year.

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A. $\exists x[P(x) \wedge R(x)]$
B. $\forall x\,[P(x) \rightarrow R(x)]$
C. $\exists x[P(x) \wedge \neg R(x)]$
D. $\forall x\,R(x)$
**Correct option: (B)**

Q9. Translate the following English sentence to FOL.

Every horse that has won a race this year is not a thoroughbred.

A. $\forall x\,[R(x) \vee Q(x)]$
B. $\forall x[R(x) \rightarrow \neg Q(x)]$
C. $\exists x[R(x) \wedge \neg Q(x)]$
D. $\forall x\,[R(x) \rightarrow Q(x)]$
**Correct option: (B)**

Q10. P(c) for some element c
$\exists x\,P(x)$

We will be using this to prove statements of the form $\exists x\,P(x)$. This is _____.

A. Universal generalization
B. Universal Instantiation
C. Existential generalization
D. Existential instantiation
**Correct option: (C)**

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